Week 3 – Workshop
Exploring & Understanding Inversion

Adelaide University

Geophysics 3A: Potential Fields & Geothermics

July 31, 2026

Preparation

Pre-reading: Course Notes Ch. 13
Lecture videos:
Notes:

Purpose

This workshop introduces the structure of inverse problems in potential-field geophysics, contrasting forward prediction with inverse estimation. Students explore the role of misfit, data weighting, and regularization, and examine how non-linearity and non-uniqueness arise in gravity modelling. Through a sequence of activities, they classify linear vs. non-linear models, reason qualitatively about non-uniqueness, and engage in hands-on inversion of a buried sphere using grid search and Gauss–Newton methods.

Learning Outcomes

- Distinguish forward and inverse problems and articulate the role of misfit, weighting, and regularization.
- Classify simple geophysical relations as linear or non-linear and propose appropriate linearizations.
- Recognize manifestations of non-uniqueness in gravity and magnetic profiles and propose discriminating data.
- Derive or confirm analytical Jacobian expressions for the buried sphere model.
- Interpret misfit surfaces, inversion paths, and the effect of noise and damping.
- Explain parameter trade-offs and resolution limitations inherent to the buried-sphere problem.

Session Flow (?? min)

Mixed groups of 4–5 with geology and physics/engineering backgrounds.

0–10 min: Warm-up

What can you learn from a shadow?

- What can you determine directly from the shadow?
- What can you estimate only if additional assumptions are made?
- What information cannot be determined uniquely?
- What prior knowledge are you using when deciding what the object might be?

How does this relate to inversion?

- A shadow does not uniquely determine the object that created it.

- Likewise, a gravity anomaly does not uniquely determine the density distribution that produced it.
- The challenge of inversion is deciding which of many acceptable models is most reasonable.

10–40 min: Mini-lecture — Forward vs Inverse, Misfit, Weighting, Challenges

- Forward vs. inverse models: $m \mapsto g(m)$ vs. $d \mapsto m$.
- Misfit: residual $r = d - g(m)$; weighted least squares $\Phi_d = \|W_d r\|_2^2$.
- Role of W_d (noise) and W_m (model constraints); damping λ.
- Challenges: non-linearity (Jacobian needed), non-uniqueness (multiple m give similar $g(m)$), ill-conditioning.

40–60 min: Activity 3.A: Testing Linearity

Two conditions must be met for linearity. Let's explore how to rigorously test for linearity before we work on other functions.

60–90 min: Activity 3.B: Linear or Non-linear?

Using the equation handout:

- Identify model parameters and classify each relation as *linear*, *non-linear*, or *linearizable*.
- If non-linear, propose a linearization (log-transform, re-parameterization, Taylor expansion/Jacobian).
- Deliverable: brief notes per group.

90–100 min: Break

100–110 min: Buried Sphere — Background

We're going to look at the gravity anomaly from a buried sphere. The forward problem is non-linear with respect to the model parameters, requiring a non-linear inverse solution. This activity is broken into two parts: a look at the formulation, and an investigation using a simple program.

- App overview: Data & Model view; Misfit view; controls for acquisition, noise, synthetic model.
- Construct a true model & generate data (freeze a single noise realization).

110–125 min: Buried Sphere — Jacobian Derivatives

1. Students derive/confirm analytical expressions for $\partial g/\partial z$, $\partial g/\partial R$, $\partial g/\partial \Delta\rho$.

125–155 min: Activity 3.C: Inversion of a Buried Sphere

With the use of `buried_sphere_gui.py`, we will examine a number of aspects of inversion, including

Grid search: Explore pairwise misfit slices and 3-D misfit surface. Identify local vs global minima.
Inversion path: Run Gauss–Newton with different λ; examine step length and stability.
Effect of noise: Increase σ; observe widening misfit basins and parameter uncertainty.
Non-uniqueness: Demonstrate parameter trade-offs (e.g. $a^3\,\Delta\rho$) and resolution ellipses.
Role of Noise: Look at how noise affects results.

155–165 min: Wrap-up

Forward models compute fields from distributions of physical properties. Inverse problems attempt to solve for the distribution of physical properties given a set of observations of a field.

Inversions come with several challenges: non-linearity, non-uniqueness, instability, and underdetermined. Non-linear functions can sometimes be linearized and if not can be solved using iterative methods. Just as multiple objects can cast the same shadow, multiple distributions of a property can produce the same geophysical field anomaly. Damping can help reduce instability and adding geologic constraints can reduce non-uniqueness.

165–175 min: Preview: Gravity Theory

This week we treated gravity primarily as data to be inverted. Next week we will begin developing gravity theory itself.

We will derive the gravity field from Newton's law of universal gravitation and examine how mass distributions create gravity anomalies. We will see how simple geological bodies such as spheres, cylinders, slabs, and contacts produce characteristic gravity signatures.

Along the way, we will connect the mathematical concepts introduced in Week 1—gradients, curvature, and potential fields—to one of the most important geophysical methods.

By the end of next week, you will be able to predict how changes in density, geometry, and observation distance affect the gravity field and begin interpreting gravity anomalies in geological terms.

Workshop Notes

Workshop Notes

Activity 3.A
Testing Linearity

Purpose

Linear functions can be solved in a single inversion step using linear least squares. This method is simpler, faster, and less computationally expensive, so when possible a linear function is preferred. In this activity, you'll learn to determine whether a function is linear or non-linear.

Learning Outcomes

By the end of this activity you should be able to:

- describe the two conditions for linearity, and
- be able to rigorously demonstrate whether a function is linear or non-linear using these tests.

Conditions for Linearity

There are two conditions that must be met for a function to be linear: homogeneity,

$$f(\alpha m_1) = \alpha f(m_1)\,,$$

and additivity

$$f(m_1 + m_2) = f(m_1) + f(m_2)\,.$$

Note: something that may take some getting used to is that our tests for linearity are with respect to model parameters that we are trying to estimate via inversion, not the experimental conditions or observation position, which is how we otherwise normally think of functions.

Box .1: Example of a linear function

For instance, $y = ax + b$ is linear with respect to the model parameters $\mathbf{m} = (a, b)$. For homogeneity,

$$\begin{aligned} y(\alpha\mathbf{m}) &= (\alpha a)x + (\alpha b) \\ &= \alpha(ax + b) \\ &= \alpha y(\mathbf{m})\,. \end{aligned}$$

For additivity, where $\mathbf{m}_1 = (a_1, b_1)$ and $\mathbf{m}_2 = (a_2, b_2)$,

$$\begin{aligned} y(\mathbf{m}_1 + \mathbf{m}_2) &= (a_1 + a_2)x + (b_1 + b_2) \\ &= (a_1 x + b_1) + (a_2 x + b_2) \\ &= y(\mathbf{m}_1) + y(\mathbf{m}_2)\,. \end{aligned}$$

Since both homogeneity and additivity are satisfied, the function is linear.

Question 1. Linear function
Show that

$$k(\mathrm{SiO_2}) = a(\mathrm{SiO_2})^2 + b(\mathrm{SiO_2}) + c$$

is linear with respect to the model parameters, $\mathbf{m} = (a, b, c)$.

Question 2. Non-linear function
Show that

$$k(T) = k_0 \left(\frac{298}{T}\right)^n$$

is non-linear with respect to the model parameters, $\mathbf{m} = (k_0, n)$.

Question 3. Why is the first equation easier to invert than the second?

Key Ideas

Linearity requires additivity and homogeneity. Homogeneity means scaling the model parameters by a factor α scales the output by the same factor, $f(\alpha m_1) = \alpha f(m_1)$; additivity means the response to a sum of two parameter sets equals the sum of their individual responses, $f(m_1 + m_2) = f(m_1) + f(m_2)$. Both conditions must hold simultaneously for a function to be linear — satisfying only one is not sufficient.

Linearity is defined with respect to the model parameters, not the independent variable. A function can be non-linear in x (e.g., a polynomial in position) but perfectly linear in the unknown coefficients (a, b, c) that we wish to recover by inversion.

Activity 3.B
Linear or Non-linear?

Purpose

This activity builds the skill of classifying geophysical forward relations as linear, non-linear, or linearizable, using a range of equations drawn from gravity, magnetics, geothermics, and physical property estimation. Correct classification determines whether a problem can be solved in a single step or requires iterative inversion.

Learning Outcomes

By the end of this activity, you should be able to:

- Identify the data, model parameters, and operator in a geophysical forward relation.
- Apply the homogeneity and additivity tests to determine whether a relation is linear in its model parameters.
- Propose a linearization strategy (log-transform, change of variables, or Taylor expansion) for non-linear relations where one exists.
- Explain the practical consequences of linearity vs. non-linearity for inversion.

1 Theory

Inversion problems can be cast as $\mathbb{A}\mathbf{m} = \mathbf{d}$ for linear problems and $A(m) = d$ for non-linear problems. Where d is the data and A is the operator or design (more on that below). The difference may appear subtle, though in reality the difference is that the linear equation can be written as a linear algebra problem as the model parameters are independent and discrete. In the non-linear formulation, the one or more model parameters are dependent upon another. As a result they cannot be fully separated unless they can be linearized.

Let's explore what A, m, and d are a bit more intuitively. The data, d, are the results of an experiment. Identifying the data are generally straightforward. The operator and model parameters can takes a bit more getting used to. The model parameters, m, are the physical or virtual properties of the system and initially unknown. Once you determine these properties, the forward model becomes a prediction. The operator or design matrix, A, can be thought of *as the experiment itself,* and is dependent upon the geometry, conditions of the experiment, and the physics. The model tells us what exists, the operator tells us how we observe it.

When we are doing a geophysical experiment with the Earth as the medium,

- **m** → the model is Earth, what it is made of and its physical properties. Changing **m** changes the Earth.
- **A** → how Earth talks to the instrument, how the Earth is sampled by the experiment, i.e., the rules of the experiment. Changing **A** changes the experiment, i.e., same Earth + different experiment → different data.
- **d** → the data is the outcome of the experiment, what we measure

In most geophysical problems, **A** is

- gravity – distances from mass elements
- magnetics – sensor positions and dip angle
- heat flow – layer thicknesses and boundary conditions
- electromagnetics – distances from conductive/resistive bodies
- seismology – ray paths or travel times

So the entries in **A** are determined by:

- positions
- distances
- times
- physical laws
- boundary conditions

Put simply, the matrix $\mathbb{A}$ encodes the physics and geometry of how the experiment samples the model, **m**, resulting in the observations, **d**.

Linearizing

Many non-linear equations can be transformed into a linear form. Doing so is a great advantage because it allows for the problem to be solved using linear least squares methods and the solution is obtained in one step. Before resorting to a non-linear inversion, ask whether one of the following approaches applies.

Logarithms for power laws

If $y = ax^b$ take the logarithm:

$$\log y = \log a + b \log$$

which is linear in $\ln x$. Examples: Gardner's relation, many petrophysical correlations, empirical scaling laws.

Logarithms for exponentials

If $y = ae^{\pm bx}$ then

$$\log y = \log a \pm bx$$

which is linear. Examples: radioactive decay, compaction models, and attenuation relations.

Change of variables

Sometimes a product of unknowns behaves as a single parameter (i.e., are inseparable). For example from the gravity of a buried sphere, $g \propto a^3 \Delta\rho$. Define:

$$M = a^3 \Delta\rho$$

then $g \propto M$, which is linear in the new parameter. This is particularly useful for demonstrating why some parameters are not independently resolvable.

Reciprocal transforms

Some rational functions become linear after taking an inverse:

$$y = \frac{1}{a + bx}$$

gives

$$\frac{1}{y} = a + bx$$

which is linear.

*N*ote: When transforming, as for many of the examples above, the data, model parameters and observation conditions all have to be transformed in order to achieve a linear form. Once the least squares solution is solved, the parameters can be transformed into normal space to express the model in its original form.

Task

For each of the following forward relations drawn from potential fields, geothermics, and physical property relations, decide whether it is (i) linear, (ii) non-linear, or (iii) non-linear but can be linearized. If non-linear, describe how you would linearize or re-parameterize it. Justify your choice with respect to the model parameters that would be inverted.

In your groups:

1. Identify the data and the experimental quantities.
2. For each equation
   - identify the quantity or quantities that would be treated as the *model parameter(s)* in an inversion.
   - Classify the mapping data → model as linear or non-linear in those parameters.
   - If non-linear, propose a linearization (e.g., log-transform, Taylor expansion/Jacobian around p_0, or a change of variables) and note any loss of identifiability.

Equations

Q1. **Gravity (Bouguer slab):** We'll see this equation again as it is used to model the gravity due to an infinite slab.

$$\Delta g = 2\pi G(\rho_s - \rho_c)H$$

Q2. **Gravity (buried sphere):** This gravity solution is one of the simplest models for a gravity anomaly. We will explore it further.

$$g(r) = \tfrac{4}{3}\pi a^3 G(\rho_s - \rho_c)\frac{z}{(r^2+z^2)^{3/2}}$$

Q3. **Thermal conductivity $k(T, X)$:** We'll use this formulation in the problem set. This equation is used to estimate thermal conductivity model for a solid solution series for a mineral.

$$k(T,X) = (a + bX + cX^2)\,(298/T)^n$$

Q4. **Thermal conductivity $k(T, P)$:** This form of thermal conductivity is often used to estimate thermal conductivity for a rock.

$$k(T,P) = \frac{k_0}{A+BT}\,(1+cP)$$

Q5. **Isostasy (Airy):** Airy isostasy is an end-member isostatic model, used to explain topographic variations due to differences in crustal thickness.

$$\Delta t = \frac{\rho_c}{\rho_m - \rho_c}\, h$$

Q6. **Isostatic gravity correction (column):** Gravity can be corrected for a mass deficit created by variations in crustal thickness via an isostatic correction.

$$\Delta g_{\text{iso}} \approx 2\pi G(\rho_m - \rho_c)\,\Delta t - 2\pi G\,\rho_c\,h$$

Q7. **Magnetics (2-D infinite dike):** The ocean floor records chaotic reversals in Earth's magnetic field. The remnant magnetic field is "frozen" in dikes produced at the mid-ocean ridge.

$$\Delta T(x) = C\left[\arctan\left(\frac{x - x_1}{z}\right) - \arctan\left(\frac{x - x_2}{z}\right)\right]$$

Q8. **Layered 1-D heat with sources:** Geotherms can largely be explained by 1-D heat flow through a layered Earth. We'll see this again when we start geothermics.

$$T_i(z) = T_{i-1} + \frac{q_{i-1}}{k_i}(z_i - z_{i-1}) - \frac{A_i}{2k_i}(z_i - z_{i-1})^2$$

Q9. **Density–porosity mix:** Sedimentary rocks include pore space (porosity) that may is filled with a fluid (air, water, brine, gas, oil, etc.).

$$\rho_{bulk} = (1 - \phi)\,\rho_{matrix} + \phi\,\rho_{fluid}$$

Q10. **Gardner's relation (density–velocity):** Many solutions in geophysics involve trying to estimate one property from another, in this case density from seismic velocity.

$$\rho = a\,v^b$$

Q11. **Gravity (discrete 2-D prisms):** While we can produce analytic equations for a variety of idealized geologic bodies with simple geometries (e.g., buried spheres as above), to produce more geologically realistic bodies we can produce structures from arbitrarily-shaped polygons.

$$g_z = 2G\Delta\rho \sum_{i=1}^{N} \left[(x_{i+1} - x_i)\log\left(\frac{r_{i+1}}{r_i}\right) + (z_{i+1} - z_i)\arctan\left(\frac{x_i z_{i+1} - x_{i+1} z_i}{x_i x_{i+1} + z_i z_{i+1}}\right)\right]$$

where

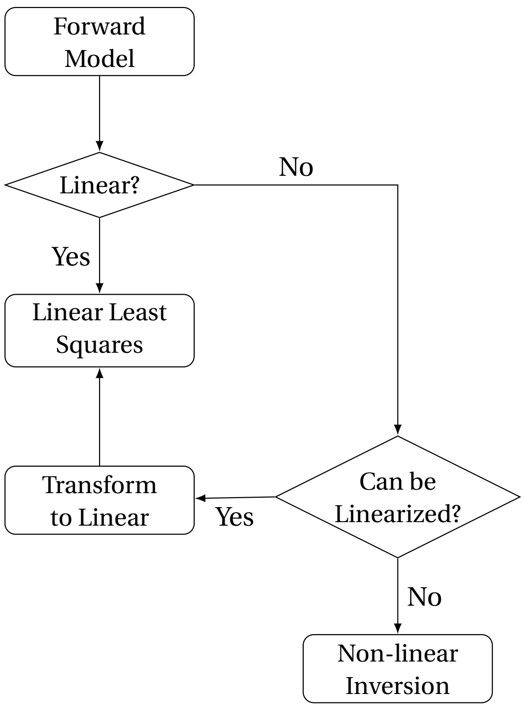
$$r_i = \sqrt{x_i^2 + z_i^2}, \qquad (x_{N+1}, z_{N+1}) = (x_1, z_1).$$

Key Ideas

The operator A encodes the physics and geometry of the experiment. Changing the Earth (the model **m**) changes the predictions; changing the experiment (the operator **A**) also changes the predictions even if the Earth stays the same.

Many geophysical relations are non-linear but can be linearized. Power laws become linear under a log-transform; functions with products of unknowns may become linear under a change of variables. Recognizing these opportunities converts a difficult iterative problem into a tractable one.

Non-linearity has consequences for inversion: the Jacobian must be recomputed at each iteration, convergence is not guaranteed, and the solution found may be a local rather than a global minimum. The choice of starting model matters far more for non-linear than for linear problems.



Activity 3.C
Inversion of a Buried Sphere

Purpose

The purpose of this activity is to develop an intuitive understanding of inversion using the gravity anomaly of a buried sphere. By comparing grid-search and gradient-based approaches, you will explore how inverse methods search model space, how model parameterization influences the solution, and why non-uniqueness and noise make geophysical interpretation challenging.

Learning Outcomes

By the end of this activity, you should be able to:

- Explain the difference between forward modelling, grid search, and gradient-based inversion.
- Describe the role of the Jacobian matrix in determining model updates.
- Identify sources of non-uniqueness in geophysical inverse problems.
- Explain how model parameterization can improve or degrade inversion performance.
- Describe the influence of noise on inversion results and uncertainty.

Box .2: Theory

Gravity of a Buried Sphere

We will return to how we derive the gravity due to a buried sphere later. For now, we will model the vertical gravity anomaly at the surface produced by a buried sphere of depth z (to center), radius a, and density contrast $\Delta\rho$. For a station at horizontal radius $r = \sqrt{x^2+y^2}$,

$$g(r|z_c, a, \Delta\rho) = \underbrace{\frac{4}{3}\pi a^3 \Delta\rho}_{\text{mass}} G \frac{z}{\left(r^2+z_c^2\right)^{3/2}}, \tag{1}$$

with G the gravitational constant.

Inverse Model

The gravity equation due to a buried sphere is non-linear, requiring a non-linear iterative solver. The code we will use employs the Gauss–Newton inverse method, which can be enhanced with Levenberg–Marquardt regularization.

We seek $\mathbf{m} = [z_c, a, \Delta\rho]^\mathsf{T}$ that minimizes the objective function,

$$\Phi(m) = \underbrace{\left\| W_d \left(\mathbf{g}_{\text{obs}} - g(\mathbf{m})\right)\right\|_2^2}_{\text{model misfit}} + \underbrace{\lambda^2 \; \|(\mathbf{m}-\mathbf{m}_0)\|_2^2}_{\text{regularization term}},$$

where $\mathbf{g}_{\text{obs}}$ are the observed gravity data, $g(\mathbf{m})$ is the predicted gravity using Equation **??**, $\mathbf{W}_d$ is a diagonal matrix of inverse variance (data uncertainty), and λ is the Levenberg–Marquart regularization parameter. Our initial guess is given by the column vector $\mathbf{m}_0$.

To determine $\mathbf{m}$, we have to solve iteratively, with each model step being

$$\Delta\mathbf{m} = (\mathbf{J}^\mathsf{T}\mathbf{W}_d^2\mathbf{J} + \lambda^2\mathbf{I})^{-1}\mathbf{J}^\mathsf{T}\mathbf{W}_d^2\left(\mathbf{g}_{\text{obs}} - g(\mathbf{m}_0)\right).$$

Box .2: continued.

Role of λ. Small λ approaches Gauss–Newton (fast, but can be unstable if $J^\top J$ is ill-conditioned). Large λ yields smaller, more stable steps (gradient-descent-like), helpful far from the solution or with strong trade-offs.

Question 1. Jacobian

Before we play with the buried sphere program, let's derive derivatives we need for the Jacobian matrix.

The Jacobian for the buried sphere requires three derivatives:

$$\mathbf{J} = \begin{bmatrix} \frac{\partial g}{\partial z_c} & \frac{\partial g}{\partial a} & \frac{\partial g}{\partial \Delta\rho} \end{bmatrix}.$$

Derive the derivatives from Equation **??**.

Exploration

The questions below relate to the `buried_sphere_gui.py` program. While the questions are about the demo, the answers are largely independent of both the problem and program.

Question 2. Grid Search

What is a grid search? Is it a forward or inverse driven solution? What does it tell us about the misfit? How do we determine the model parameters?

Question 3. Inversion

How does inversion approach the problem differently from the grid search? What path does it take to the answer?

Question 4. Comparison

How does the result of the inversion compare with that of the grid search? If the two solutions do not match, why?

Question 5. Non-uniqueness

The problem we have posed is non-unique even though we use a single "true" model to generate the data. Why is the solution non-unique?

Question 6. 2D versus 3D

Let's switch from 3 parameters to 2. The 2 parameter result is a better constrained, why? How is this improvement in constraint reflected in the misfit surfaces?

Question 7. Noise

How does noise affect the solution via inversion? How does noise contribute to non-uniqueness?

For this demonstration, repeat the inversion with 0%, 5%, and 10% Gaussian noise.

Key Ideas

Grid search is a brute-force forward modelling approach to systematically sample the model space with misfit is computed at every combination of model parameters. It can identify the global minimum within the explored model domain, but it becomes computationally expensive as the number of model parameters increases and requires a sufficiently dense sampling of the model space to accurately resolve the misfit surface. For many large difficult problems found in geophysics, a grid search is unfeasible.

Inversion uses information about the local shape of the misfit surface, typically through the Jacobian matrix, to move iteratively toward a low-misfit solution. While each iteration is computationally more expensive than a single forward calculation, inversion generally requires far fewer model evaluations than a grid search.

Best of all, inversion remains practical for geophysical problems with many parameters (tens to millions). The trade-off is that inversion explores only a small portion of model space and may converge to a local minimum rather than the global minimum.

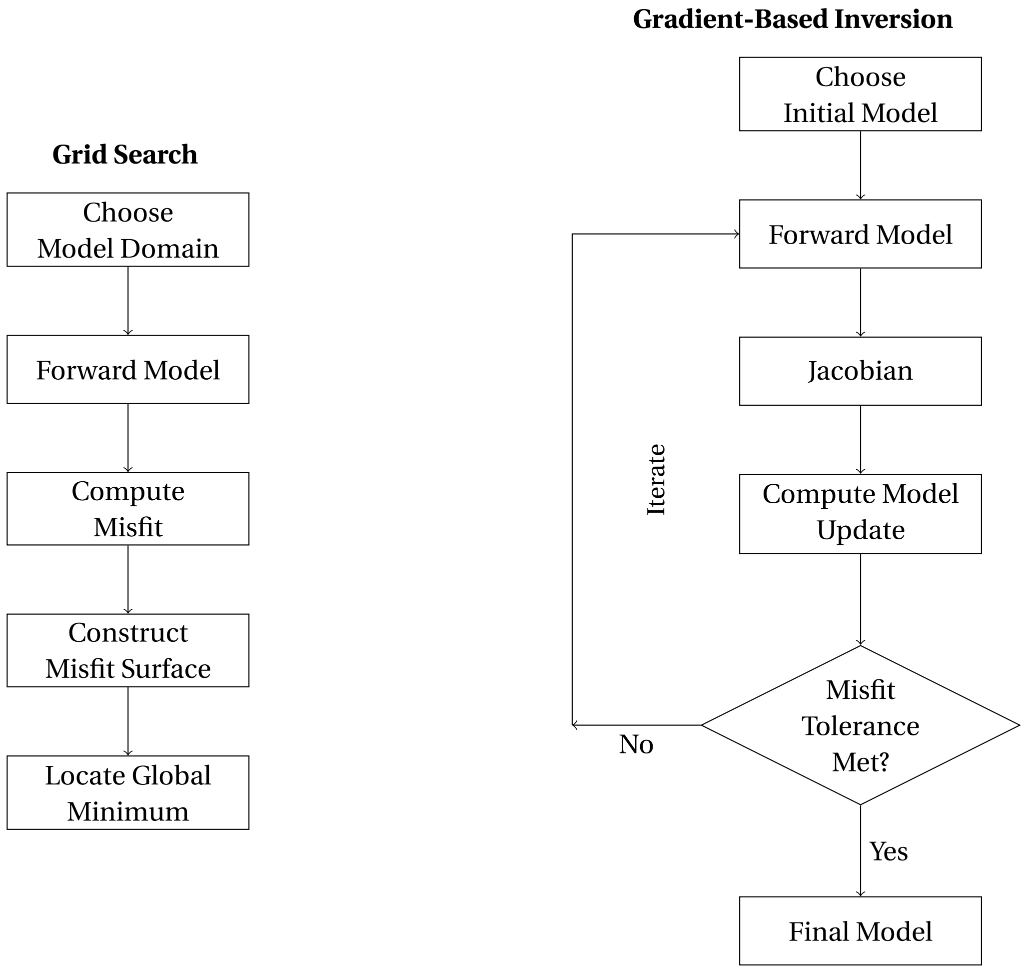


Figure 1: Comparison of grid-search and gradient-based inversion approaches.